Spring 2026 MSML606 Extra Credit Project 2
(deadlines: 4/15 for proposal and 5/06 for final submission)

Overview

Apply **graph algorithms or dynamic programming** to a real-world dataset (e.g., topological sort, shortest path, BFS/DFS for graph algorithms, or LCS, TSP, 0/1 knapsack for DP). You may also choose an NP-hard or NP-complete problem and apply exact or approximate methods.

We will be using the same policy as for the first bonus project. The bonus credit will be applied to the final exam score.

Instructions:

This assignment adheres to all course policies specified in the syllabus, with the following project-specific modifications:

- ➢ Do not use AI to "start" your logic or generate your initial approach
- ➢ You may use AI tools to assist in generating the User Interface (UI) of your application
- ➢ Originality requirement: Other than the UI components, all code, comments, and documentation (README) must be authored and typed manually by you.

Timeline & requirements

Proposal submission (deadline 4/15):

- ➢ Submit a max 1-2 page document or slide deck (ELMS)
- ➢ Include team composition (individual or groups of up to 3)
- ➢ Specify the problem, dataset, and brief description of the algorithm and its usage

Final Submission (deadline 5/06):

- ➢ Share on ELMS (ELMS->discussion->Project2):
  - ○ Post a 3-minute video demonstration of your project
  - ○ Respond to at least 2 other posts
- ➢ GitHub repository
  - ○ Source code and setup instructions (how to run the example)
  - ○ Project website (if applicable)
  - ○ Include your original proposal and presentation files in the repo.

An opportunity to present your work may be offered if time permits.

Target audience:

The final project (GitHub and Video) must be designed for a non-CS audience. The goal is for any reader without a CS background to understand exactly what the problem is, and why your algorithm can be applied to solve it without spending effort beyond watching your video or reading your documentation.

Rubric: (Maximum Score: 23 Marks)

Problem	Criteria	Marks
Proposal (due 4/15)	- Submitted on time, correct format with 1-2 pages/slides (1 marks) - Team composition clearly stated (1 marks) - Dataset is identified and described (2 marks) - Application idea is clear and uses the chosen algorithm meaningfully (2 marks)	6
Code & GitHub Repository (due 5/06)	- Chosen algorithm is correctly implemented and central to the problem (2 marks) - Code is functional (i.e. runs without errors per setup instructions) (2 marks) - Instructions are clear and code is readable and well-commented (2 marks) - Proposal and presentation files included in repo (1 marks) - Evaluation based on commit history, code organization, and clarity of documentation (2 marks)	9
Video Presentation (due 5/06)	- Demonstrates all core features of the application (2 marks) - Accessible to a non-CS audience with no unexplained jargon; Target audience awareness across all submissions (2 marks) - Within 3-minute time limit (1 marks)	5
Peer Engagement (due 5/06)	- Responded to at least 2 classmates posts (1 marks) - Responses are substantive, not just "great job!" (1 marks)	2
Originality & Academic Integrity	- State of using AI/external sources (must have) - No AI used for core logic or initial approach - Code, comments, and README are manually authored	Must have
Creativity	Originality in the application or chosen dataset	1

NB: Problem difficulty and UI aesthetics are not evaluated!